

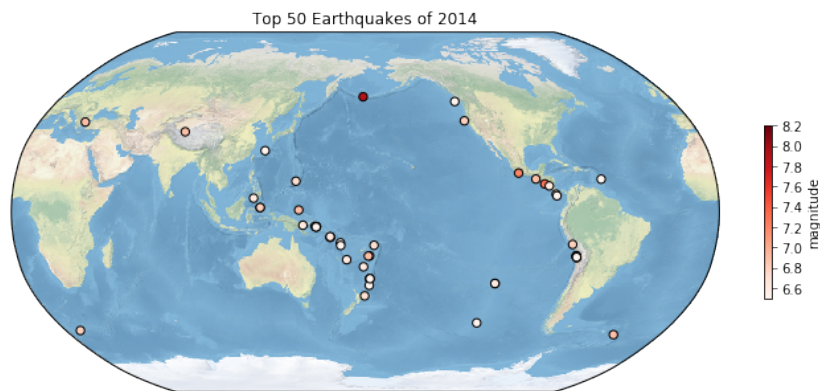
Assignment 04

环境准备：

```
1 import numpy as np
2 import xarray as xr
3 import pandas as pd
4 import matplotlib.pyplot as plt
5 import matplotlib as mpl
6 import cartopy.crs as ccrs
7 import cartopy.feature as cfeature
8 import geopandas as gpd
```

1. Global Earthquakes

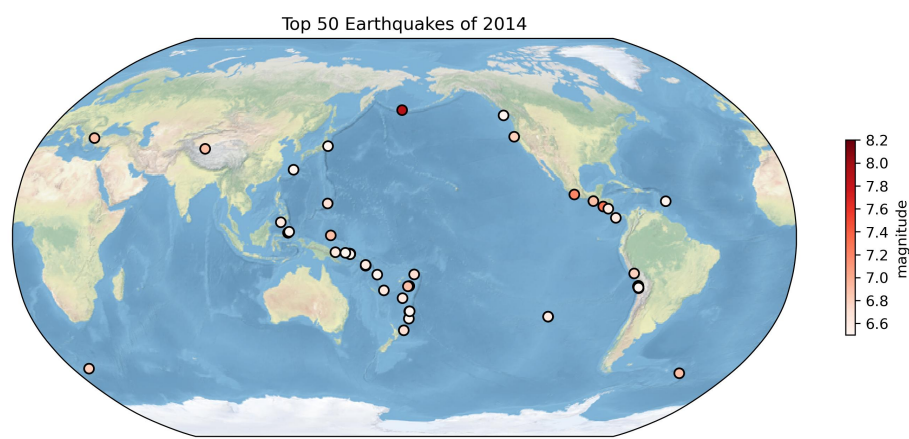
In this problem set, we will use [this file](#) from the USGS Earthquakes Database. The dataset is similar to the one you use in [Assignment 02](#). Use the file provided (usgs_earthquakes.csv) to recreate the following map. Use the mag column for magnitude. [10 points]



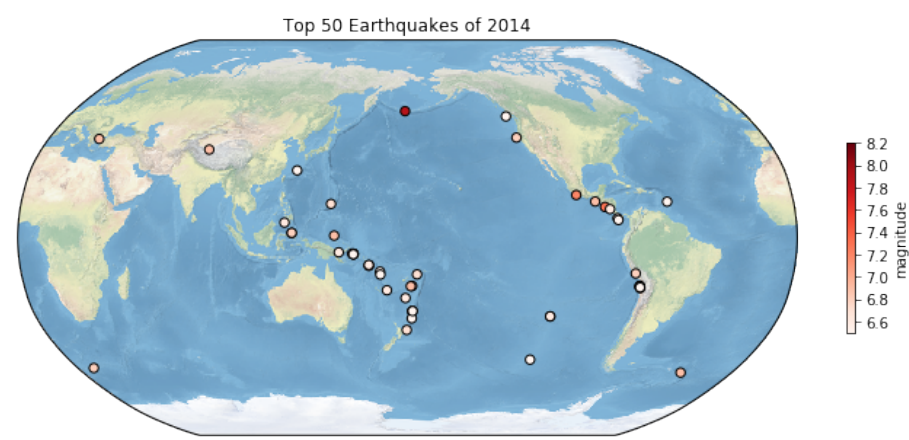
读取相关数据库，根据时间筛选地震数据，并提取震级排名前 50 的经纬度和震级数据。依照参考图设置图像尺寸、投影信息、绘图类型、颜色主题、线条粗细、标记大小、底图填充、标题及色标等。

```
1 df = pd.read_csv("../data/usgs_earthquakes.csv", parse_dates=["time"])
2 df = df.loc[pd.DatetimeIndex(df["time"]).year==2014]
3 df = df.sort_values(by="mag", ascending=False).iloc[0:50]
4
5 fig = plt.figure(figsize=np.array([8, 4])*1.5, dpi=100)
6 spec = fig.add_gridspec(ncols=1, nrows=1)
7 ax = fig.add_subplot(spec[0], projection=ccrs.Robinson(central_longitude=-180))
8 ax.set_extent([0.1, 359.9, -90, 90])
9 ar = ax.scatter(df["longitude"], df["latitude"], s=45, lw=1.2, c=df["mag"],
10               cmap="Reds", ec="k", transform=ccrs.PlateCarree())
11 ax.stock_img()
12 ax.set_title("Top 50 Earthquakes of 2014")
13 fig.colorbar(ar, ax=ax, label="magnitude", ticks=np.arange(6.6, 8.3, 0.2),
14             shrink=0.4)
15 fig.savefig("../images/PS4_1_figure1.jpg", bbox_inches="tight", dpi=300)
```

最终得到绘图结果如下：



题目版本：



2. Explore a netCDF dataset

Browse the NASA's Goddard Earth Sciences Data and Information Services Center (GES DISC) [website](#). Search and download a dataset you are interested in. You are also welcome to use data from your group in this problem set. But the dataset should be in netCDF format. For this problem set, you are welcome to use the same dataset you used in [Assignment 03](#).

这里选择数据为 [TROPOMI 的月均全球 0.1° NO2 观测数据](#)（上一次作业所使用的数据）。考虑到数据量相对单个仓库较为庞大（>2GB），这里仅保留最终用于绘图的数据：

- 所有时段平均值：

HAQ_TROPOMI_NO2_GLOBAL_QA75_L3_Monthly_Mean_V2.4_201901_to_202508.nc

下列代码读取用于绘图的基本数据、feature 等信息：

```
1 # data for Figure
2 ds =
  xr.open_dataset("./data/HAQ_TROPOMI_NO2_GLOBAL_QA75_L3_Monthly_Mean_V2.4_201901_to_20
  2508.nc")
3 ds_select_china = ds.sel(lon=slice(65, 140), lat=slice(10, 60))
4
5 # cartopy feature
6 gdf_world = gpd.read_file("./data/world.zh.json")
7 gdf_china = gpd.read_file("./data/china_province.geojson").sort_values(by="gb")
8 geojson_world = gdf_world["geometry"]
9 geojson_china = gdf_china["geometry"]
10 gdf_china = gdf_china.loc[gdf_china["name"]!="境界线"]
11 gdf_china.reset_index(inplace=True)
12 shp_world = cfeature.ShapelyFeature(geojson_world, ccrs.PlateCarree(), fc="none",
  ec="k", lw=0.5)
13 shp_china = cfeature.ShapelyFeature(geojson_china, ccrs.PlateCarree(), fc="none",
  ec="k", lw=0.5)
```

2.1 [10 points]

Make a global map of a certain variable. Your figure should contain: a project, x label and ticks, y label and ticks, title, gridlines, legend, colorbar, masks or features, annotations, and text box (1 point each).

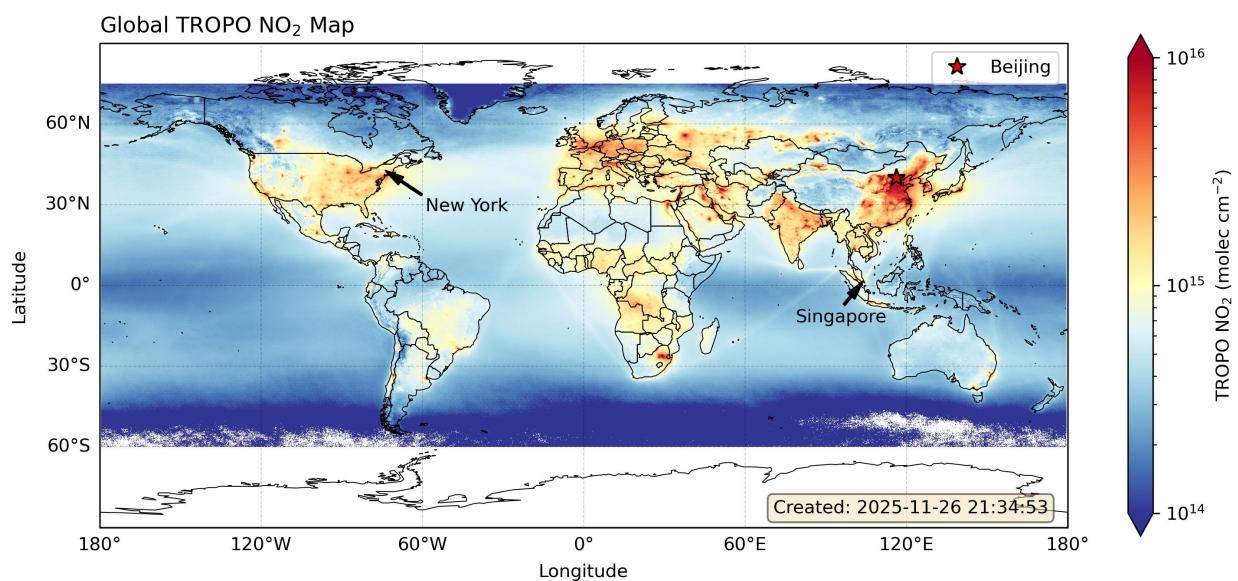
依照题目要求，绘制了全球对流层 NO2 均值如下，并增加相关标记，使用经纬度投影绘制。

```
1 import matplotlib.lines as mlines
2 blue_line = mlines.Line2D([], [], color='blue', marker='*',
3                             markersize=15, label='Blue stars')
4
5 fig = plt.figure(figsize=[12, 5], dpi=100)
6 spec = fig.add_gridspec(ncols=1, nrows=1)
7
8 # PlateCarree Project
9 ax = fig.add_subplot(spec[0], projection=ccrs.PlateCarree())
10 ax.set_extent([-180, 180, -90, 90])
```

```

11 # Map
12 ar = ax.pcolormesh(ds["lon"], ds["lat"], ds["Tropospheric_NO2"],
norm=mpl.colors.LogNorm(vmin=1e14, vmax=1e16), cmap="RdYlBu_r",
transform=ccrs.PlateCarree())
13 # X-label, Y-label
14 ax.set_xlabel('Longitude', labelpad=20)
15 ax.set_ylabel('Latitude', labelpad=40)
16 ax.set(xticks=[], yticks=[])
17 # Title
18 ax.set_title(r"Global TROPO NO2 Map", loc="left")
19 # Gridlines, X-ticks, Y-ticks
20 gl = ax.gridlines(color="k", linestyle="--", lw=0.5, draw_labels=True,
x_inline=False, y_inline=False, dms=True, alpha=0.2)
21 gl.top_labels, gl.right_labels, gl.rotate_labels = False, False, False
22 # Legend
23 ax.plot([116.40753], [39.90403], "r*", mec="k", ms=10, label="Beijing",
transform=ccrs.PlateCarree())
24 ax.legend(loc='upper right', bbox_to_anchor=(1, 1))
25 # Colorbar
26 fig.colorbar(ar, ax=ax, extend="both", label=r"TROPO NO2 (molec cm-2)")
27 # Masks or Features
28 ax.add_feature(shp_world)
29 # Annotations
30 ax.annotate("New York", xy=(-73.754968, 42.6511674), xytext=(-73.754968+15,
42.6511674-15), arrowprops=dict(facecolor='black', shrink=0.01, width=1, headwidth=5,
headlength=10), transform=ccrs.PlateCarree())
31 ax.annotate("Singapore", xy=(103.822872, 1.364917), xytext=(103.822872-25,
1.364917-15), arrowprops=dict(facecolor='black', shrink=0.01, width=1, headwidth=5,
headlength=10), transform=ccrs.PlateCarree())
32 # Text Box
33 ax.text(0.98, 0.03, "Created: 2025-11-26 21:34:53", transform=ax.transAxes,
horizontalalignment="right", bbox=dict(boxstyle='round', facecolor='wheat',
alpha=0.5))
34 fig.savefig("./images/PS4_2_figure1.jpg", bbox_inches="tight", dpi=300)

```



2.2 [10 points]

Make a regional map of the same variable. Your figure should contain: a different project, x label and ticks, y label and ticks, title, gridlines, legend, colorbar, masks or features, annotations, and text box (1 point each).

依照题目要求，绘制了中国大陆及周边对流层 NO₂ 均值如下，并增加相关标记，使用兰伯特投影绘制。

```
1  fig = plt.figure(figsize=[6, 3], dpi=100)
2  spec = fig.add_gridspec(ncols=1, nrows=1)
3  # Lambert Project
4  ax = fig.add_subplot(spec[0], projection=ccrs.LambertConformal(central_longitude=120,
5  central_latitude=34, standard_parallels=[20, 40]))
6  ax.set_extent([82, 132, 16, 51])
7  # Map
8  ar = ax.contourf(ds_select_china["lon"], ds_select_china["lat"],
9  ds_select_china["Tropospheric_NO2"], levels=20, cmap="Reds",
10 transform=ccrs.PlateCarree())
11 # X-label, Y-label
12 ax.set_xlabel('Longitude', labelpad=20)
13 ax.set_ylabel('Latitude', labelpad=40)
14 ax.set(xticks=[], yticks=[])
15 # Title
16 ax.set_title(r"China TROPO NO2 Map", loc="left")
17 # Gridlines, X-ticks and Y-ticks
18 gl = ax.gridlines(color="k", linestyle="--", lw=0.5, draw_labels=["bottom", "left"],
19 x_inline=False, y_inline=False, dms=True, alpha=0.2, rotate_labels=False)
20 # Legend
21 ax.plot([116.40753], [39.90403], "r*", mec="k", ms=10, label="Beijing",
22 transform=ccrs.PlateCarree())
23 ax.legend(loc='upper right', bbox_to_anchor=(1, 1))
24 # Colorbar
25 fig.colorbar(ar, ax=ax, extend="both", label=r"TROPO NO2 (molec cm-2)")
26 # Masks or Features
27 ax.add_feature(shp_china)
28 ax.coastlines()
29 # Annotates
30 ax.annotate("Beijing", xy=(116.4, 39.9), xytext=(116.4-20, 39.9+5),
31 arrowprops=dict(facecolor='black', shrink=0.01, width=1, headwidth=5, headlength=10),
32 transform=ccrs.PlateCarree())
33 ax.annotate("Shenzhen", xy=(114.085947, 22.547), xytext=(114.085947+5, 22.547-3),
34 arrowprops=dict(facecolor='black', shrink=0.01, width=1, headwidth=5, headlength=10),
35 transform=ccrs.PlateCarree())
36 # Text Box
37 ax.text(0.03, 0.05, "Created: 2025-12-09", transform=ax.transAxes,
38 horizontalalignment="left", bbox=dict(boxstyle='round', facecolor='wheat',
39 alpha=0.5))
40 fig.savefig("./images/PS4_2_figure2.jpg", bbox_inches="tight", dpi=300)
```

